# Bird Song Embedding Visualization

Progress report: what has been built, what the embeddings show, and what comes next

3 October 2026 · sub-project of *BEATs: Audio Pre-Training with Acoustic Tokenizers*

## Summary


- Bird recordings are turned into **768-number vectors** (embeddings) by a pretrained BEATs audio model, and loaded into the TensorFlow Embedding Projector to explore them in 3D.
- The corpus is **20 Texas Gulf Coast bird species** from iNaturalist: **12,007 five-second clips** from **3,958 recordings** before quality filtering.
- Off-the-shelf BEATs already groups birds by species: 25.5% of each clip's 10 nearest neighbours are the same species, against about 5% by chance (**4.9× chance**). Real structure, but far from clean.
- Data quality mattered more than any modelling choice so far. A frequency-band filter removes wind and traffic noise and lifts purity to 32–33%, but at a 1.5 kHz cutoff it **destroys the two dove species** (White-winged Dove falls *below* chance).
- Lowering the cutoff to **400 Hz** (`band400`) gives the best result: **5.8× chance**, 45% more clips than the strict filter, and the doves recover. It is the corpus chosen for the next phase.
- Next: train a bird-specific model (Phase 3b) on band400 and compare it with these baselines in the Projector.


## 1. Where the project stands

| Phase | State | Notes |
|---|---|---|
| 1 — Data acquisition | Done (iNaturalist) | ~5,100 recordings, 20 species. Xeno-canto, BirdCLEF and field-recording fetchers are built and each waits on a credential or the audio itself. |
| 2 — Preprocessing | Done | 16 kHz mono, silence-trimmed, 5 s clips, split 70/15/15 by recording. |
| 3a — Baseline embeddings | Done | BEATs `iter3_plus_AS2M`, frozen, 768-d. Five filter variants embedded. |
| 3b — Bird-pretrained model | Not started | Corpus chosen (band400). GPU sized on an RTX A4000: batch 64 fits in 12.8 GiB. |
| 4 — Projector export | Done | baseline, filtered and band400 exported and validated. |
| 5 — Visualization & analysis | Done for 3a | kNN purity per species for every variant. |
| 6 — Documentation | Ongoing | README, FINDINGS, REPRODUCE, ATTRIBUTION, this report. |

## 2. The data

All audio is research-grade iNaturalist recordings of 20 common Gulf Coast species, released under five Creative Commons licences (full attribution in `exports/ATTRIBUTION.md`). Species codes are eBird's 6-letter codes, e.g. `normoc` = **Nor**thern **Moc**kingbird. Each recording contributes at most 8 clips, so one long recording cannot dominate its species, and with 20 species a random neighbour is the same species only about 5% of the time.

| Species | Code | Clips (baseline) | Recordings | Clips (band400) |
|---|---|---|---|---|
| Northern Mockingbird *Mimus polyglottos* | normoc | 924 | 206 | 556 |
| Great-tailed Grackle *Quiscalus mexicanus* | grtgra | 774 | 249 | 375 |
| Northern Cardinal *Cardinalis cardinalis* | norcar | 742 | 236 | 338 |
| Red-winged Blackbird *Agelaius phoeniceus* | rewbla | 670 | 220 | 251 |
| Eastern Bluebird *Sialia sialis* | easblu | 669 | 211 | 235 |
| Carolina Wren *Thryothorus ludovicianus* | carwre | 653 | 220 | 408 |
| American Robin *Turdus migratorius* | amerob | 651 | 199 | 312 |
| Downy Woodpecker *Dryobates pubescens* | dowwoo | 648 | 197 | 275 |

| Species | Code | Clips (baseline) | Recordings | Clips (band400) |
|---|---|---|---|---|
| White-winged Dove *Zenaida asiatica* | whwdov | 615 | 217 | 272 |
| Mourning Dove *Zenaida macroura* | moudov | 580 | 176 | 200 |
| Blue Jay *Cyanocitta cristata* | blujay | 580 | 194 | 309 |
| House Sparrow *Passer domesticus* | houspa | 564 | 205 | 311 |
| Barn Swallow *Hirundo rustica* | barswa | 544 | 212 | 241 |
| Tufted Titmouse *Baeolophus bicolor* | tuftit | 516 | 182 | 244 |
| Carolina Chickadee *Poecile carolinensis* | carchi | 507 | 184 | 232 |
| Killdeer *Charadrius vociferus* | killde | 500 | 184 | 178 |
| Great Blue Heron *Ardea herodias* | grbher | 498 | 178 | 234 |
| American Crow *Corvus brachyrhynchos* | amecro | 489 | 167 | 220 |
| Red-bellied Woodpecker *Melanerpes carolinus* | rebwoo | 458 | 172 | 168 |
| Laughing Gull *Leucophaeus atricilla* | laugul | 425 | 149 | 132 |
| **Total** | | **12,007** | **3,958** | **5,491** |

## 3. How a clip becomes a point

Every 5-second clip becomes exactly **one vector of 768 numbers**, and each vector is one point in the Projector. The shapes below were confirmed on the GPU benchmark (`scripts/bench_gpu.py`).

| Step | Shape |
|---|---|
| Waveform: 5 s at 16 kHz | (batch, 80,000 samples) |
| Log-mel spectrogram: 128 bands, 10 ms hop | (batch, 498 frames, 128 bands) |
| Cut into 16×16 patches: 31 in time × 8 in frequency | 248 patches |
| BEATs encoder output | (batch, 248, 768) |
| Average over the 248 patches | **(batch, 768)** |

The average is taken over time *and* frequency together, because BEATs stores its patches as one flat sequence. The code comment says “mean over time”, which is slightly inaccurate.

### The quality filters (chosen by us, applied before the model)

None of these settings is learned. They decide which clips the model is shown. The model is the only learned step, and the purity scores are computed afterwards to grade the vectors.

| Setting | Value | What it removes |
|---|---|---|
| Silence trim (relative) | 35 dB below the recording's peak | Dead air at the start and end of each recording. |
| Minimum length | 3 s after trimming | Recordings too short for one clip. |
| Loudness floor (absolute) | RMS ≥ 0.0035 (≈ −49 dBFS) | Near-silent 5 s windows, e.g. pauses between songs. |
| Band filter | ≥ 25% of energy in [floor] – 9 kHz | Loud clips that are not birdsong: wind, traffic, handling noise. |

The trim scales with each recording; the loudness floor is fixed. For quiet recordings the floor therefore does most of the filtering, and no rationale for the value 0.0035 is recorded anywhere in the project.

### Measuring quality: kNN species purity

For each clip, find its 10 nearest neighbours in the full 768-dimensional space, ignoring other clips from the same recording (those are near-copies and would inflate the score). Purity is the share of those neighbours that are the same species, averaged over all clips. It is judged against **chance**, the purity random neighbours would give; the ratio is the **lift**.

## 4. Findings

Five versions of the corpus were embedded with the same model. They differ only in the quality filter. `baseline` and `filtered` come from the 8 September corpus; the other three from a 20 September re-fetch, which reproduced the original numbers to within 2%.

| Export | Filter | Clips | Purity | Chance | Lift | Mourning / White-winged Dove |
|---|---|---|---|---|---|---|
| baseline | loudness only | 12,007 | 25.5% | 5.2% | 4.9× | 26.2% / 30.7% |
| nofilter | loudness only (re-run) | 11,757 | 26.2% | 5.2% | 5.0× | 23.6% / 30.8% |
| **band400** | **+ band 400 Hz–9 kHz** | **5,491** | **32.3%** | **5.6%** | **5.8×** | **33.0% / 42.8%** |
| filtered | + band 1.5–9 kHz | 3,809 | 32.3% | 6.2% | 5.2× | 10.6% / 6.5% |
| band1500 | + band 1.5–9 kHz (re-run) | 3,774 | 33.0% | 6.2% | 5.3× | 10.3% / 4.6% |

### 4.1 Off-the-shelf BEATs carries real species structure, but not enough

At about 5× chance, a model never trained on birds clearly encodes species-relevant sound. But the best species reaches 63–68% purity while the worst sit near chance. This is the number a bird-pretrained model has to beat.

### 4.2 Data quality dominated everything

- **Lost .m4a files.** The audio library could not decode iNaturalist's .m4a files, silently dropping 64% of recordings. Decoding them through ffmpeg took the corpus from 1,390 to 12,007 clips.
- **Loud is not the same as bird.** A spot-check found 8 of 20 clips that passed the loudness floor were wind or traffic rumble below 500 Hz. The band filter brought that to 0 of 20 and raised purity by about 7 points.

### 4.3 A 1.5 kHz cutoff destroys low-pitched species; 400 Hz fixes it

Doves coo at roughly 300–700 Hz, below a 1.5 kHz cutoff, so that filter throws away their actual song and keeps only odd clips. White-winged Dove falls to 4.6%, **below its 6.2% chance rate**. Downy Woodpecker (low-frequency drumming) is also hurt: 18.1% unfiltered to 9.4%. With a 400 Hz cutoff, White-winged Dove recovers to 42.8% (4th of 20) and Mourning Dove to 33.0%.

band400 improves 18 of 20 species over no band filter. The exception is Eastern Bluebird (14.0% → 9.4%), so 400 Hz is a better single cutoff, not a perfect one; a per-species band would be the principled fix.

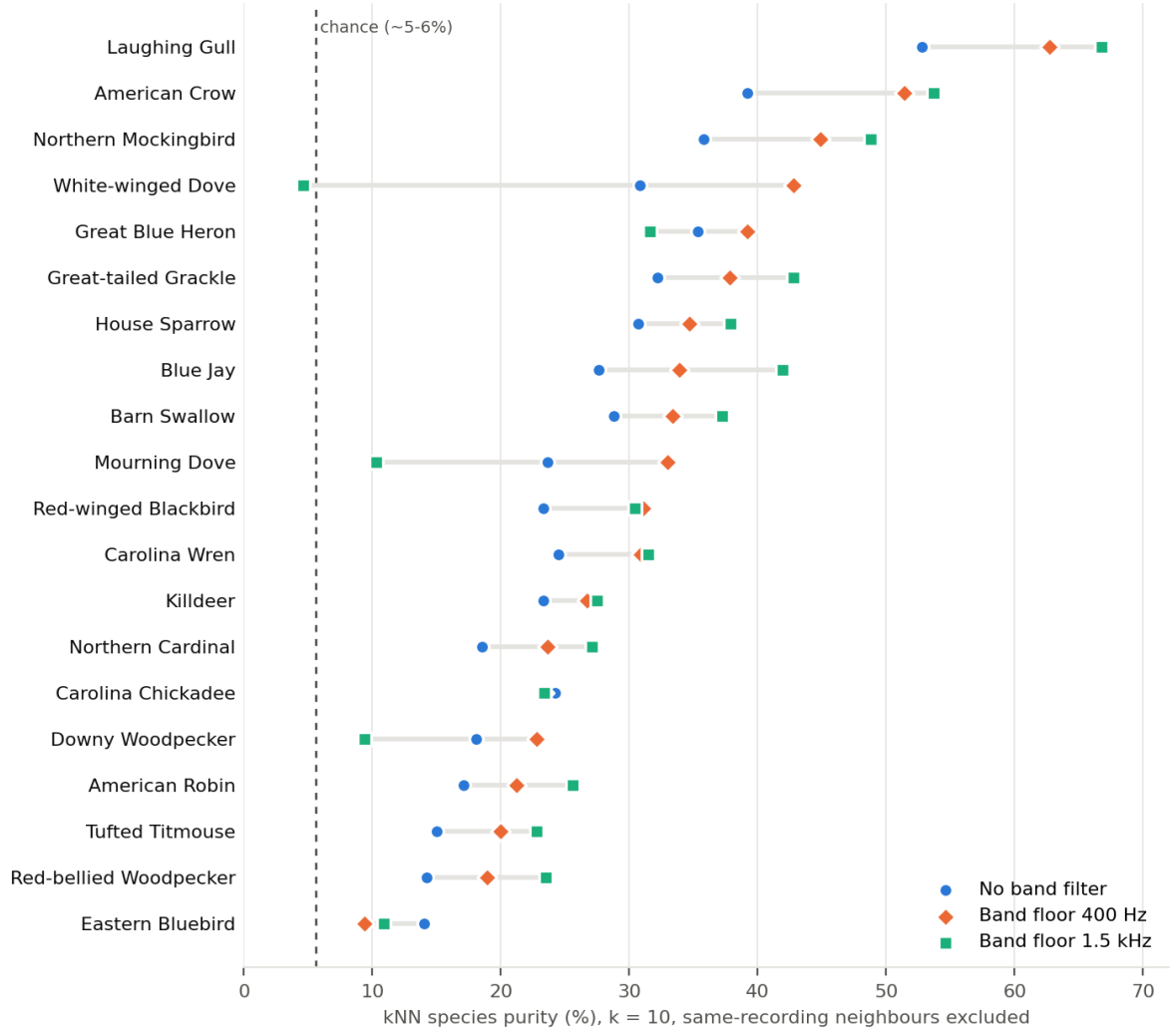



Figure 1. Purity per species under the three 20 September filters, sorted by band400. The grey bar spans the three values. The doves and Downy Woodpecker collapse at 1.5 kHz; band400 keeps them.

### 4.4 The species ordering makes ecological sense

Laughing Gull is the tightest cluster in every run (53–68%): a loud, broadband, very stereotyped call. American Crow and Great-tailed Grackle also rank high. The most overlapped are quiet, tonal, narrow-band singers such as Tufted Titmouse, Eastern Bluebird and Red-bellied Woodpecker, which is what you would expect from a model trained on general audio.

### 4.5 Not yet answerable: source bias

All audio is from iNaturalist, so we cannot yet test whether clusters reflect *where the audio came from* rather than *which bird is singing*. The check is built and will run automatically once a second source is added.

## 5. Visualizing in the Embedding Projector

Each export is two tab-separated files matched by line number: line *i* of `metadata.tsv` describes the vector on line *i* of `vectors.tsv`. A validation script checks widths, NaN/Inf values, line endings, row counts and row order.

| Export | Points × dims | vectors.tsv | Status |
|---|---|---|---|
| baseline | 12,007 × 768 | 87.6 MB | exported, validated |
| filtered | 3,809 × 768 | 27.8 MB | exported, validated |
| band400 | 5,491 × 768 | 40.1 MB | exported, validated (3 Oct) |
| nofilter, band1500 | — | — | embeddings exist; not yet exported |

To view: open projector.tensorflow.org → Load → vectors.tsv, then metadata.tsv. Set **Label by** to common_name and **Color by** to species_code. Files are read in the browser and nothing is uploaded.

### Reading the 3D view

- **PCA** (default) keeps the directions of greatest variance. The first 10 components hold only ~39% of the variance, so species overlap heavily in 3D. Use it for a quick look and for spotting batch effects.
- **t-SNE / UMAP** keep each point's local neighbourhood and show clusters far better, but distances between clusters and cluster sizes mean little. The browser uses a 10,000-point sample, so band400 and filtered fit completely; baseline does not.
- The 3D picture is for exploring. The purity numbers and the Projector's nearest-neighbour panel both use the full 768 dimensions and are the reliable measures.

## 6. Known issues

- **Unreadable default labels.** The first metadata column is clip_id (e.g. `inaturalist_inat100360457s366439_seg02`), which the Projector uses as the label. Workaround: Label by common_name.
- **Messy country field.** Derived from free-text place names: 228 distinct values, mixing US / USA / United States and including park names.
- **Broken accented characters** (e.g. México arrives as “M?xico”) from a text-encoding step in the iNaturalist fetch.
- **Undocumented loudness floor.** 0.0035 RMS was never tuned, and it is measured on raw gain, so quiet but genuine recordings lose the most clips.
- **Single source.** Only iNaturalist, so source bias cannot be tested yet.
- **Local copies.** The baseline and filtered .npy embeddings and all clip WAVs are only in Azure blob storage, not on this machine.

## 7. Next steps

- **Phase 3b:** train the bird-pretrained model on band400 (RTX A4000, batch ≤ 64), export it as `birdpretrained`, and compare it with baseline and band400 side by side.
- **Clean the metadata:** a readable first column, ISO country codes, and fixed text encoding. This is export-only, so the embeddings stay the same.
- **Better Projector views:** spectrogram thumbnails per point (sprite), a recording_id column, and a per-point purity column to colour confused regions.
- **Measure the 3D distortion:** what share of each point's 768-D neighbours survive in PCA, t-SNE and UMAP.
- **Add sources:** a Xeno-canto API v3 key and a Kaggle token unlock two more datasets and the source-bias test; field recordings drop into the same species list.
- **Revisit the filters:** a recording-relative loudness floor and per-species frequency bands.